

ON ARITHMETICAL STRUCTURES WITH A MULTI-EDGE

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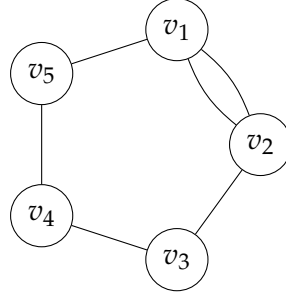
1. INTRODUCTION

Let $n > 1$ and $k > 1$ be arbitrary integers. A *tilde-cycle* is a connected multigraph (without self-loops) on n vertices $v_1, \dots, v_n$ denoted $\tilde{\mathcal{C}}_{n,k}$ where the degrees $\deg(v_1) = \deg(v_2) = k + 1$ and $\deg(v_i) = 2$ for all $2 < i \leq n$. Thus, we view a tilde-cycle as the pair $\tilde{\mathcal{C}}_{n,k} = (V, E)$ where V and E denote the vertex set and edge set of $\tilde{\mathcal{C}}_{n,k}$ respectively.

Definition 1.1. Suppose that A is the adjacency matrix of $\tilde{\mathcal{C}}_{n,k}$. Then an *arithmetical structure* on $\tilde{\mathcal{C}}_{n,k}$ is a pair of vectors $(\mathbf{d}, \mathbf{r}) \in (\mathbb{Z}_{>0})^n \times (\mathbb{Z}_{>0})^n$ such that $\mathbf{r}$ is primitive and $(\text{diag}(\mathbf{d}) - A)\mathbf{r} = \mathbf{0}$.

2. ARITHMETICAL STRUCTURES ON CYCLES WITH A MULTI-EDGE

Recall that an arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ is a pair $(\mathbf{d}, \mathbf{r}) \in (\mathbb{Z}_{>0})^n \times (\mathbb{Z}_{>0})^n$ such that $\mathbf{r}$ is primitive and $(\text{diag}(\mathbf{d}) - A)\mathbf{r} = \mathbf{0}$, where A is the adjacency matrix of $\tilde{\mathcal{C}}_{n,k}$. Equivalently, arithmetical structures on $\tilde{\mathcal{C}}_{n,k}$ can be defined as positive integer labels on the vertex set such that for every vertex

FIGURE 1. The graph $\tilde{C}_{5,2}$

$v \in \tilde{C}_{n,k}$ and every other vertex in its neighborhood $N(v) = \{u_1, \dots, u_t\}$ satisfies

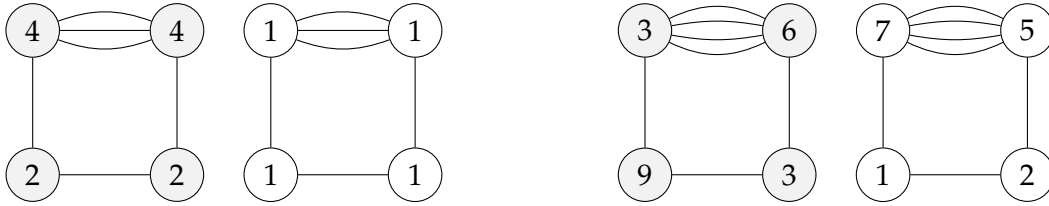
$$\mathbf{d}(v)\mathbf{r}(v) = \mathbf{r}(u_1) + \dots + \mathbf{r}(u_t).$$

More explicitly, the set of equations that define an arithmetical structure on $\tilde{C}_{n,k}$ are

$$d_i \cdot r_i = \begin{cases} r_2 k + r_n & \text{if } i = 1 \\ r_1 k + r_3 & \text{if } i = 2 \\ r_{i-1} + r_{i+1} & \text{if } 2 < i \leq n, \end{cases} \quad (1)$$

where $r_{n+1} = r_1$.

Example 2.1. Consider the arithmetical structures $(\mathbf{d}, \mathbf{r}) = ((4, 4, 2, 2), (1, 1, 1, 1))$ on $\tilde{C}_{4,3}$ and $(\mathbf{d}', \mathbf{r}') = ((3, 6, 3, 9), (7, 5, 2, 1))$ on $\tilde{C}_{4,4}$.

FIGURE 2. Arithmetical structures $(\mathbf{d}, \mathbf{r})$ on $\tilde{C}_{4,3}$ and $\tilde{C}_{4,4}$ respectively.

Let $\text{Arith}(\tilde{C}_{n,k}) = \{(\mathbf{d}, \mathbf{r}) \mid (\mathbf{d}, \mathbf{r}) \text{ is an arithmetical structure on } \tilde{C}_{n,k}\}$. We now present a condition that the $\mathbf{r}$ vector needs to satisfy in an arithmetical structure on $\tilde{C}_{n,k}$.

Proposition 2.1. *If $(\mathbf{d}, \mathbf{r}) \in \text{Arith}(\tilde{C}_{n,k})$ then*

$$\gcd(r_1, r_2) = \gcd(r_1, k) = \gcd(r_2, k) = 1.$$

Proof. Assume $(\mathbf{d}, \mathbf{r}) \in \text{Arith}(\tilde{C}_{n,k})$. Take the equation $r_3 = -r_1 k + d_2 r_2$ and let $g = \gcd(r_1, r_2)$. Then by the latter equation $\gcd(r_1, r_2, r_3) = g$. Choose any

integer i with $2 < i \leq n$ and assume inductively that $\gcd(r_{i-1}, r_i) = g$. Take the equation $r_{i+1} = -d_i r_i + r_{i-1}$. Observe that $\gcd(r_{i-1}, r_i) = g$, so induction on i yields that $\gcd(r_i, r_{i+1}) = g$ for all $1 \leq i \leq n$. By our initial assumption $\mathbf{r}$ is primitive, thus $g = 1$. Now we will show that $\gcd(r_1, k) = \gcd(r_2, k) = 1$. Take the equation $r_n = d_1 r_1 - r_2 k$ then $\gcd(r_1, k) = \gcd(r_1, r_n) = 1$ and by symmetry of $\tilde{\mathcal{C}}_{n,k}$ $\gcd(r_2, k) = 1$. $\square$

Remark 1. A converse of Proposition 2.1 is introduced on section 4.1 to generate tilde-cycles using a division algorithm.

Proposition 2.2. *Let $(\mathbf{d}, \mathbf{r}) = ((d_1, d_2, \dots, d_n), (r_1, r_2, \dots, r_n)) \in \text{Arith}(\tilde{\mathcal{C}}_{n,k})$ then*

$$(\mathbf{d}', \mathbf{r}') = ((d_2, d_1, d_n, d_{n-1}, \dots, d_3), (r_2, r_1, r_n, r_{n-1}, \dots, r_3)) \in \text{Arith}(\tilde{\mathcal{C}}_{n,k}).$$

Moreover, if $(\mathbf{d}, \mathbf{r})$ is not the Laplacian, then $(\mathbf{d}, \mathbf{r}) \neq (\mathbf{d}', \mathbf{r}')$.

Proof. Let $G \cong \tilde{\mathcal{C}}_{n,k}$ and consider the arithmetical graphs $(G, \mathbf{r})$ and $(G, \mathbf{r}')$. Suppose that there exists an automorphism $\phi : V(G) \rightarrow V(G)$ such that $r'_{\phi(v)} = r_v$ for all $v \in V(G)$. Then ϕ induces an isomorphism between arithmetical graphs. In particular, because ϕ preserves the adjacency relation in G then the divisibility conditions are preserved. $\square$

2.1. Subdivision and Smoothing. Let $(\mathbf{d}, \mathbf{r})$ be in $\text{Arith}(\tilde{\mathcal{C}}_{n,k})$ and choose any edge contained in $E(\tilde{\mathcal{C}}_{n,k})$ with endpoints v_j and v_{j+1} where $1 < j \leq n$. Then we can construct a new tilde-cycle on $n + 1$ vertices with a structure $(\mathbf{d}', \mathbf{r}')$ by removing the edge $v_j v_{j+1}$ from $\tilde{\mathcal{C}}_{n,k}$ and adding a new vertex v_{n+1} connected to v_j and v_{j+1} . We name this operation *subdivision* and define $\mathbf{r}'$ by

$$r'_i = \begin{cases} r_i & \text{for } i \in \{1, 2, \dots, n\}, \\ \mathbf{r}(v_j) + \mathbf{r}(v_{j+1}) & \text{for } i = n + 1, \end{cases} \quad (2)$$

and define $\mathbf{d}'$ by

$$d'_i = \begin{cases} d_i & \text{for } i \in \{1, 2, \dots, n\} \setminus \{j, j+1\}, \\ d_i + 1 & \text{for } i \in \{j, j+1\}, \\ 1 & \text{for } i = n + 1. \end{cases} \quad (3)$$

Then $(\mathbf{d}', \mathbf{r}')$ gives an arithmetical structure. Now we define *smoothing*, the reverse operation of subdivision. That is, given a vertex of degree 2 where the d -value is 1, then that vertex can be removed, its two neighbors connected

by an edge, and the d -value of the neighbors decreased by 1 to obtain an arithmetical structure on a tilde cycle with $n - 1$ vertices. Then $\mathbf{r}'$ is given by

$$r'_i = \begin{cases} r_i(k+1) & \text{for } i = 1, \\ r_x + r_y & \text{for } i = n+1, \end{cases} \quad (4)$$

2.2. Reverse Keyes-Reiter operation. Let $(\mathbf{d}, \mathbf{r})$ be in $\text{Arith}(\tilde{\mathcal{C}}_{n,k})$ with $\mathbf{d} = (d_1, \dots, d_n)$ and $\mathbf{r} = (r_1, \dots, r_n)$. Then we can obtain a new structure $(\mathbf{d}', \mathbf{r}')$ by applying the reverse of the *Keyes-Reiter* operation. Define $\mathbf{r}'$ by

$$r'_i = \begin{cases} r_1 + r_2 k & \text{if } i = 1, \\ r_i & \text{if } i \in \{2, 3, \dots, n\}, \\ r_1 & \text{if } i = n+1, \end{cases} \quad (5)$$

and define $\mathbf{d}'$ by

$$d'_i = \begin{cases} 1 & \text{if } i = 1, \\ 2d_2 + 1 & \text{if } i = 2, \\ d_i & \text{if } i \in \{3, 4, \dots, n\}, \\ d_1 + 1 & \text{if } i = n+1. \end{cases} \quad (6)$$

3. ENUMERATION

We now proceed to study the enumeration of arithmetical structures on $\tilde{\mathcal{C}}_{n,k}$.

Definition 3.1. Suppose that $(\mathbf{d}, \mathbf{r}) \in \text{Arith}(\tilde{\mathcal{C}}_{n,k})$. Then $(\mathbf{d}, \mathbf{r})$ is a *smooth* arithmetical structure if $d_i > 1$ for all $2 < i \leq n$. We denote by $\text{SArith}(\tilde{\mathcal{C}}_{n,k})$ the set of all smooth arithmetical structures on $\tilde{\mathcal{C}}_{n,k}$.

Observe that our definition does not restrict d_1 or d_2 . If $(\mathbf{d}, \mathbf{r}) \in \text{SArith}(\tilde{\mathcal{C}}_{n,k})$ and $d_i > 1$ for all $1 \leq i \leq n$ then we say that $(\mathbf{d}, \mathbf{r})$ is a *strongly* smooth arithmetical structure. Otherwise, if $d_1 = 1$ or $d_2 = 1$ and $d_i > 1$ for all $2 < i \leq n$ we say that $(\mathbf{d}, \mathbf{r})$ is a *weakly* smooth arithmetical structure. The set of strongly smooth (resp. weakly smooth) arithmetical structures will be denoted $\text{SSArith}(\tilde{\mathcal{C}}_{n,k})$ (resp. $\text{WSArith}(\tilde{\mathcal{C}}_{n,k})$). By definition

$$\text{SArith}(\tilde{\mathcal{C}}_{n,k}) = \text{SSArith}(\tilde{\mathcal{C}}_{n,k}) \cup \text{WSArith}(\tilde{\mathcal{C}}_{n,k}).$$

Example 3.1. From Example 2.1, the pair $(\mathbf{d}', \mathbf{r}')$ is a strongly smooth arithmetical structure and the Laplacian structure $(\mathbf{d}, \mathbf{r})$ is not smooth.

Proposition 3.1. *Let $n \geq 2$, then*

$$|\text{Arith}(\tilde{\mathcal{C}}_{n,k})| = \sum_{m=2}^n B(n-2, n-m) |\text{SArith}(\tilde{\mathcal{C}}_{m,k})|.$$

Proof. [Alexander will add a reference...](#) □

Lemma 3.1. *Let $(\mathbf{d}, \mathbf{r}) \in \text{SSArith}(\tilde{\mathcal{C}}_{n,k})$ and $f_1, f_2, \dots, f_n, g_1, g_2, \dots, g_n$ be reverse Keyes-Reiter operations. If*

$$f_1 f_2 \dots f_n((\mathbf{d}, \mathbf{r})) = g_1 g_2 \dots g_n((\mathbf{d}, \mathbf{r}))$$

then $f_i = g_i$ for $i \in \{1, 2, \dots, n\}$.

Proof. We proceed by induction on n . If $n = 1$ suppose $f_1((\mathbf{d}, \mathbf{r})) = g_1((\mathbf{d}, \mathbf{r}))$...

Suppose that □

Proposition 3.2. *Let $n \geq 2$, then*

$$|\text{WSArith}(\tilde{\mathcal{C}}_{n,k})| = \sum_{j=1}^{n-1} 2^{n-j} |\text{SSArith}(\tilde{\mathcal{C}}_{j,k})|.$$

Proof. □

Theorem 3.1. *Let $n \geq ?$, then*

$$|\text{Arith}(\tilde{\mathcal{C}}_{n,k})| = \sum_{m=2}^n \sum_{j=1}^m B(n-2, n-m) * 2^{m-j} |\text{SSArith}(\tilde{\mathcal{C}}_{j,k})|.$$

Proof. From Proposition 3.1, we get

$$|\text{Arith}(\tilde{\mathcal{C}}_{n,k})| = \sum_{m=2}^n B(n-2, n-m) |\text{SArith}(\tilde{\mathcal{C}}_{m,k})|.$$

Since $\text{SArith}(\tilde{\mathcal{C}}_{n,k})$ is the disjoint union of $\text{SSArith}(\tilde{\mathcal{C}}_{n,k})$ and $\text{WSArith}(\tilde{\mathcal{C}}_{n,k})$, we get

$$|\text{Arith}(\tilde{\mathcal{C}}_{n,k})| = \sum_{m=2}^n B(n-2, n-m) * (|\text{WSArith}(\tilde{\mathcal{C}}_{m,k})| + |\text{SSArith}(\tilde{\mathcal{C}}_{m,k})|)$$

and by substituting Proposition 3.2, we get

$$\begin{aligned}
& \sum_{m=2}^n B(n-2, n-m) * \left(\sum_{j=1}^{n-1} 2^{n-j} |\text{SSArith}(\tilde{\mathcal{C}}_{j,k})| + |\text{SSArith}(\tilde{\mathcal{C}}_{m,k})| \right) \\
&= \sum_{m=2}^n B(n-2, n-m) * \sum_{j=1}^m 2^{m-j} |\text{SSArith}(\tilde{\mathcal{C}}_{j,k})| \\
&= \sum_{m=2}^n \sum_{j=1}^m B(n-2, n-m) * 2^{m-j} |\text{SSArith}(\tilde{\mathcal{C}}_{j,k})|. \quad \square
\end{aligned}$$

4. STRONGLY SMOOTH ARITHMETICAL STRUCTURES ON CYCLES WITH A MULTI-EDGE

Motivated by the result in Theorem 3.1, in this section we will study strongly smooth arithmetical structures on $\tilde{\mathcal{C}}_{n,k}$. Recall that an arithmetical structure $(\mathbf{d}, \mathbf{r})$ is strongly smooth if no entry on the $\mathbf{d}$ vector is 1. By Proposition 2.2, we can split all non-Laplacian arithmetical structures on $\tilde{\mathcal{C}}_{n,k}$ into two equal size sets depending on whether $r_1 > r_2$ or $r_2 > r_1$. For simplicity, we will assume in this section that $r_1 \geq r_2$.

Proposition 4.1. *Let $(\mathbf{d}, \mathbf{r})$ be a non-Laplacian strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ with $r_1 > r_2$, then $r_1 > r_i$ for all $i \in \{2, 3, \dots, n\}$.*

Proof. Suppose by contradiction that r_1 is not the maximum value in $\mathbf{r}$. Let $r_j = \max\{r_i \mid i \in \{1, 2, \dots, n\}\}$. Note that $j \neq 1$ by assumption and $j \neq 2$ since $r_1 > r_2$. Since r_j must divide $r_{j-1} + r_{j+1}$ (where $r_{n+1} = r_1$) and it is greater than each of r_{j-1} and r_{j+1} then $r_j = r_{j-1} + r_{j+1}$. Therefore, $d_j = 1$, which contradicts that $(\mathbf{d}, \mathbf{r})$ is a strongly smooth arithmetical structure. $\square$

Proposition 4.2. *Let $(\mathbf{d}, \mathbf{r})$ be a non-Laplacian strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ with $r_1 > r_2$, then $d_1 \in \{2, 3, \dots, k\}$.*

Proof. Since $(\mathbf{d}, \mathbf{r})$ is a strongly smooth arithmetical structure, $d_1 > 1$. By assumption $r_1 > r_2$ and by Proposition 4.1, $r_1 > r_n$, thus

$$d_1 = \frac{kr_2 + r_n}{r_1} = \frac{kr_2}{r_1} + \frac{r_n}{r_1} < k + 1. \quad \square$$

Proposition 4.3. *If $(\mathbf{d}, \mathbf{r}) \in \text{SSArith}(\tilde{\mathcal{C}}_{n,k})$ and $r_1 > r_2$ then $d_1 = \left\lceil \frac{r_2 k}{r_1} \right\rceil$.*

Proof. Use that $r_1 > r_n$ $\square$

Check this proof below.

Proposition 4.4. *Let $(\mathbf{d}, \mathbf{r})$ be a non-Laplacian strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ with $r_1 > r_2$, then $d_2 > \left\lceil \frac{k^2}{d_1} \right\rceil$ when $\frac{k^2}{d_1}$ is not an integer, and $d_2 > \frac{k^2}{d_1} + 1$ when $\frac{k^2}{d_1}$ is an integer.*

Proof. Let $r_1 = \frac{kr_2+r_n}{d_1}$ and $d_2 = \frac{kr_1+r_3}{r_2}$. Then,

$$d_2 = \frac{k(\frac{kr_2+r_n}{d_1}) + r_3}{r_2} = \frac{\frac{k^2r_2+kr_n}{d_1} + r_3}{r_2} = \frac{k^2r_2 + r_n + d_1r_3}{d_1r_2} = \frac{k^2}{d_1} + \frac{kr_n + d_1r_3}{d_1r_2}.$$

Note that in the case where $\frac{k^2}{d_1}$ is an integer, $\frac{kr_n+d_1r_3}{d_1r_2}$ is an integer greater than 1.

□

Proof. In order to evaluate the upper bound of d_2 , we split the known equation into four cases:

(1) Case 1: $r_3 < r_2, r_n < r_2$.

If $d_2 = \frac{k^2}{d_1} + \frac{kr_n}{d_1r_2} + \frac{r_3}{r_2}$ then $\frac{kr_n}{d_1r_2} = \frac{k}{d_1} * \frac{r_n}{r_2}$ where $\frac{r_n}{r_2} < 1$ and $\frac{r_3}{r_2} < 1$.

This means that $d_2 < \frac{k^2}{d_1} * \frac{k}{d_1}(1) + 1 = \frac{k^2+k+d_1}{d_1}$.

So $d_2 < \frac{k^2+k+d_1}{d_1}$ when $r_3 < r_2$ and $r_n < r_2$.

(2) Case 2: $r_3 < r_2, r_n \geq r_2$.

If $d_2 = \frac{k^2}{d_1} + \frac{kr_n+d_1r_3}{d_1r_2}$ then $\frac{kr_n}{d_1r_2} = \frac{k}{d_1} * \frac{r_n}{r_2}$ where $\frac{r_n}{r_2} \geq 1$ and $\frac{r_3}{r_2} < 1$

but $r_n < r_1$ and $r_1 = \frac{kr_2+r_n}{d_1}$, so

$$r_n < \frac{kr_2 + r_n}{d_1}$$

$$d_1r_n < kr_2 + r_n$$

$$r_n < kr_2$$

. Then, when $r_n = kr_2$,

$$d_2 < \frac{k^2}{d_1} + \frac{k}{d_1} * \frac{kr_2}{r_2} + \frac{r_3}{r_2}$$

$$d_2 < \frac{k^2}{d_1} + \frac{k^2r_2}{d_1r_2} + \frac{r_3}{r_2}$$

$$d_2 < \frac{2k^2}{d_1} + \frac{r_3}{r_2}$$

where $\frac{r_3}{r_2} < 1$, so $d_2 < \frac{2k^2}{d_1} + 1$ when $r_3 < r_2$ and $r_n \geq r_2$.

(3) Case 3: $r_3 > r_2, r_n \leq r_2$. The structure is not smooth

(4) Case 4: $r_3 > r_2, r_n > r_2$.

We know that in most cases $r_2 > r_3$, but when $r_3 > r_2$, then [more to say here but I can add it](#) $r_{n+1} > r_n$ in order to keep the structure smooth. With this rule, $r_n \not\leq r_2$. So this case does not exist.

In the case where $d_1 = k$, we are able to get a stronger upper bound using divisibility conditions. We will proceed using a proof by contradiction. Thus, $d_2 \geq k + 3$. Using divisibility conditions, we know that $d_n = \frac{r_1 + r_{n-1}}{r_n}$. Therefore, $d_n r_n = r_1 + r_{n-1}$. Because we are assuming strong smoothness, we can rewrite this equation as an inequality, where $2r_n \leq r_1 + r_{n-1}$. This can also be represented as $r_n - r_{n-1} \leq r_1 - r_n$. If you keep using these divisibility equations, you get an inequality that looks like

$$r_1 - r_n \geq r_n - r_{n-1} \geq r_{n-1} - r_{n-2} \geq \cdots \geq r_4 - r_3 \geq r_3 - r_2.$$

Using divisibility conditions we know that, $d_2 r_2 = kr_1 + r_3$. Solving for $r_3 - r_2$, we get

$$r_3 - r_2 = (d_2 - 1)r_2 - kr_1 \geq (k + 2)r_2 - kr_1.$$

Using the divisibility condition $d_1 = \frac{kr_2 + r_n}{r_1}$, we get $kr_1 = kr_2 + r_n$, we know that $r_2 = \frac{kr_1 - r_n}{k}$. Plugging this into $(k + 2)r_2 - kr_1$, we see that

$$(k + 2)r_2 - kr_1 = k + 2\left(\frac{kr_1 - r_n}{k}\right) - kr_1 = \frac{k^2 r_1 - kr_n}{k} + \frac{2kr_1 - 2r_n}{k} + r_1 - r_n - kr_1 \leq r_3 - r_2$$

. Therefore, $r_1 - r_n \geq 2r_1 - \frac{5}{3}r_n$, which simplifies to $-r_1 - \frac{8}{3}r_n \geq 0$, a contradiction. Thus, when $d_1 = 3, d_2 \leq 5$. $\square$

Is 5—234 the only structure that is increasing?

4.1. Generating strongly smooth structures.

Definition 4.1. Let a_1 and a_2 be positive integers. We define the sequence $F(a_1, a_2) = (a_1, \dots, a_N)$ by letting a_{i+1} be the least residue of $-a_{i-1} \pmod{a_i}$ for each $i \geq 2$ and N is the largest index such that $a_N > 0$.

Example 4.1. For (a_1, a_2) choose $(13, 6)$ and $(20, 14)$. Then we have the sequences $F(13, 6) = (13, 6, 5, 4, 3, 2, 1)$ and $F(20, 14) = (20, 14, 8, 2)$

Lemma 4.1. If $F(a_1, a_2) = (a_1, a_2, \dots, a_N)$ then $a_N = \gcd(a_1, a_2)$.

Proof. Let $x, y \in \mathbb{Z}$. Because $\gcd(x, y) = \gcd(-x, y)$ then $F(-x, y)$ is as the Euclidean algorithm. $\square$

Proposition 4.5. We can construct a pair $(\mathbf{d}, \mathbf{r}) \in \text{Arith } n, k$ if and only if r_1, r_2 , and k are pair-wise coprime.

Proof. Assume that $(\mathbf{d}, \mathbf{r}) \in \text{Arith } n, k$. Then by Proposition 2.1 it follows that $\gcd(r_1, r_2) = 1$. Let $g = \gcd(r_1, k)$ and $g' = \gcd(r_2, k)$ and take the equation $r_n = -d_1 r_1 + r_2 k$. Because r_n is a linear combination of g then $g \mid r_n$. Choose any integer i with $1 < i < n$ and assume inductively that $\gcd(r_i, r_{i-1}) = g$. Take the equation $r_{i+1} = -r_{i-1} + d_i r_i$. Because r_{i+1} is a linear combination of g then $g \mid r_{i+1}$. Therefore, induction on i shows that $g \mid r_1, \dots, r_n$. Apply the same argument for g' . Because $\mathbf{r}$ is primitive then $g = g' = 1$ so r_1, r_2 , and k are pair-wise coprime.

Conversely, assume that r_1, r_2 , and k are pair-wise coprime. Define the sequences $F(r_2 k, r_1) = (a_1, \dots, a_N)$ and $F(r_1 k, r_2) = (b_1, \dots, b_M)$ where N, M are the largest indexes such that $a_N, b_M > 0$. Let G be the connected graph with vertex set $\{a_2, \dots, a_N\} \cup \{b_2, \dots, b_M\}$ such that $\deg(a_2) = \deg(b_2) = k + 1$ and $G[a_2, \dots, a_N] + G[b_2, \dots, b_M] \cong P_N + P_M$.



FIGURE 3. A drawing of G .

We will show that the graph $G + a_N b_M$ is in $\text{Arith}(\tilde{\mathcal{C}}_{n,k})$. Recall our assumption and it follows that $\gcd(r_2 k, r_1) = \gcd(r_1 k, r_2) = 1$. Apply Lemma 6.1 and get that $a_N = b_M = 1$. Merging vertex a_N and b_M yields a tilde cycle of the form $\tilde{\mathcal{C}}_{n,k}$ where $n = N + M - 1$.

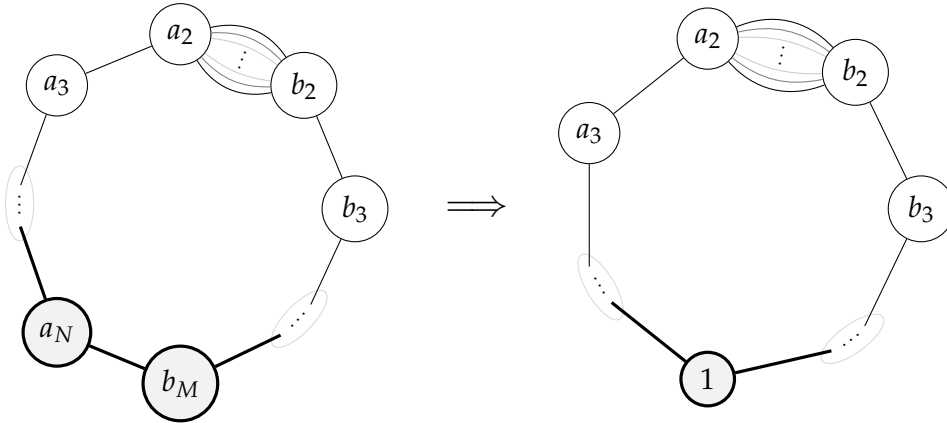


FIGURE 4. The resulting tilde-cycle on $N + M - 1$ vertices by merging a_N and b_M .

Our proof is complete by setting $\mathbf{r} = (a_2, b_2, \dots, b_{M-1}, 1, a_{N-1}, \dots, a_3)$ for which $\mathbf{d}$ is uniquely determined. $\square$

Corollary 4.1. *If $(\mathbf{d}, \mathbf{r}) \in \text{Arith}(\tilde{\mathcal{C}}_{n,k})$ then $d_1 \geq \left\lceil \frac{r_2 k}{r_1} \right\rceil$ and $d_2 \geq \left\lceil \frac{r_1 k}{r_2} \right\rceil$.*

Proof. Assume $(\mathbf{d}, \mathbf{r}) \in \text{Arith}(\tilde{\mathcal{C}}_{n,k})$, so $d_1 = \frac{r_2 k + r_n}{r_1}$. By Lemma 4.5, r_1, r_2 and k are pairwise coprime, so $\frac{r_2 k}{r_1} \notin \mathbb{Z}$. Since $\frac{r_2 k + r_n}{r_1} \in \mathbb{Z}$, then

$$\left\lceil \frac{r_2 k}{r_1} \right\rceil \leq \frac{r_2 k}{r_1} + \frac{r_n}{r_1} = d_1.$$

By symmetry, we get the desired bound for d_2 . $\square$

4.2. Push operation on strongly smooth arithmetical structures for a general case. The push operation is a operation performed on strongly smooth arithmetical structures on $\mathcal{C}_{n,k}$ graphs with $d_1 = k, d_2 = k + 1$ and $d_3, d_4, \dots, d_n \neq 1$.

Definition 4.2. Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ with $d_1 = k$. Let $\mathbf{r}'$ and $\mathbf{d}'$ be the vectors defined as:

$$r'_i = \begin{cases} r_1 + r_n & \text{if } i = 1 \\ r_2 + r_n & \text{if } i = 2 \\ r_{i-1} & \text{if } i \in \{3, 4, \dots, n+1\} \end{cases} \quad \text{and } d'_i = \begin{cases} k & \text{if } i = 1 \\ k + 1 & \text{if } i = 2 \\ d_2 - 2 & \text{if } i = 3 \\ d_i & \text{if } i \in \{4, 5, \dots, n\} \\ d_n + 1 & \text{if } i = n + 1 \end{cases}$$

Lemma 4.2. *Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure with $d_1 = 3$ and $(\mathbf{d}', \mathbf{r}')$ be as in Definition 4.2. Then, $(\mathbf{d}', \mathbf{r}')$ is an arithmetical structure on $\tilde{\mathcal{C}}_{n+1,k}$.*

Proof. Since $r'_i = r_{i-1}$ for $i \in \{3, 4, \dots, n+1\}$ then divisibility conditions at r'_i for $i \in \{4, 5, \dots, n\}$ follow from the divisibility conditions on the structure $(\mathbf{d}, \mathbf{r})$. Note that since $d_1 = k$ and $d_1 r_1 = k r_2 + r_n$, then $k r_1 = k r_2 + r_n$ and $k r_2 = k r_1 - r_n$. For the remaining divisibility conditions, we know that d'_1 can be written as:

$$\begin{aligned} d'_1 &= \frac{k r'_2 + r'_n}{r'_1} = \frac{k(r_2 + r_n) + r_n}{r_1 + r_n} = \frac{k r_2 + (k + 1) r_n}{r_1 + r_n} = \\ &= \frac{(k r_1 - r_n) + (k + 1) r_n}{r_1 + r_n} = \frac{k r_1 + k r_n}{r_1 + r_n} = k. \end{aligned}$$

For d'_2 , we know

$$d'_2 = \frac{k r'_1 + r'_3}{r'_2} = \frac{k r_1 + k r_n + r_2}{r_2 + r_n} = \frac{k r_2 + r_n + k r_n + r_2}{r_2 + r_n} =$$

$$\frac{(k+1)r_2 + (k+1)r_n}{r_2 + r_n} = k+1.$$

And for d'_3 ,

$$\begin{aligned} d'_3 &= \frac{r'_2 + r'_4}{r'_3} = \frac{r_2 + r_n + r_3}{r_2} = 1 + \frac{r_n + r_3}{r_2} = \\ 1 + \frac{kr_1 - kr_2 + r_3}{r_2} &= 1 - k + \frac{kr_1 + r_3}{r_2} = 1 - k + d_2. \end{aligned}$$

Lastly, for d'_{n+1} ,

$$d'_{n+1} = \frac{r'_1 + r'_n}{r'_{n+1}} = \frac{r_1 + r_n + r_{n-1}}{r_n} = \frac{r_1 + r_{n-1}}{r_n} + 1 = d_n + 1.$$

□

Definition 4.3. Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,3}$ with $d_1 = 3$. and $d_2 = 4$ The reverse of push operation has $\mathbf{r}'$ and $\mathbf{d}'$ that are defined as:

$$r'_i = \begin{cases} r_1 - r_n & \text{if } i = 1 \\ r_2 - r_n & \text{if } i = 2 \\ r_{i+1} & \text{if } i \in \{3, 4, \dots, n-1\} \end{cases} \quad \text{and } d'_i = \begin{cases} k & \text{if } i = 1 \\ k - 1 + d_3 & \text{if } i = 2 \\ d_{i+1} & \text{if } i \in \{3, 4, \dots, n-2\} \\ d_n - 1 & \text{if } i = n-1 \end{cases}$$

Lemma 4.3. *The reverse of the push operation will give you a proper arithmetical structure.*

Proof. Because the only r values changing are the r values at $r'_1, r'_2, r'_3, r'_{n-1}$, we must check that the corresponding d values remain integers. We see that d'_1 is an integer since

$$\begin{aligned} d'_1 &= \frac{kr'_2 + r'_{n-1}}{r'_1} = \frac{k(r_2 - r_n) + r_n}{r_1 - r_n} = \frac{kr_2 - (k-1)r_n}{r_1 - r_n} = \frac{k(\frac{kr_1 - r_n}{k}) - (k-1)r_n}{r_1 - r_n} = \\ &= \frac{kr_1 - r_n - (k-1)r_n}{r_1 - r_n} = \frac{k(r_1 - r_n)}{r_1 - r_n} = k. \end{aligned}$$

For d'_2 , note that $d_1 = k$, $d_2 = k+1$, $(k+1)r_2 = kr_1 + r_3$ and $kr_1 - kr_2 = r_n$ so $r_2 - r_n = r_2 - (kr_1 - kr_2) = -kr_1 + (k+1)r_2 = -kr_1 + kr_1 + r_3 = r_3$. This can be used to determine that

$$d'_2 = \frac{kr'_1 + r'_3}{r'_2} = \frac{kr_1 - kr_n + r_4}{r_2 - r_n} = \frac{kr_2 - (k-1)r_n + r_4}{r_2 - r_n} =$$

$$\frac{(k-1)r_2 - (k-1)r_n + r_2 + r_4}{r_2 - r_n} = (k-1) + \frac{r_2 + r_4}{r_2 - r_n} = (k-1) + \frac{r_2 + r_4}{r_3} = k-1 + d_3$$

And d'_3 is an integer since

$$d'_3 = \frac{r'_2 + r'_4}{r'_3} = \frac{r_2 - r_n + r_5}{r_4} = \frac{r_3 + r_5}{r_4} = d_4$$

This works for d'_{n-1} as well since

$$d'_{n-1} = \frac{r'_1 + r'_{n-2}}{r'_{n-1}} = \frac{r_1 - r_n + r_{n-1}}{r_n} = \frac{r_1 + r_{n-1}}{r_n} - 1 = d_n - 1.$$

□

4.3. Flip operation on strongly smooth arithmetical structures. The flip operation is an operation performed on strongly smooth arithmetical structures where $d_1 = k$ and $d_2 = k + 2$.

Definition 4.4. Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,k}$ with $d_1 = k$ and $d_2 = k + 2$. Let $\mathbf{r}'$ and $\mathbf{d}'$ be the vectors defined as

$$r'_i = \begin{cases} r_1 + r_n - r_2 & \text{if } i = 1 \\ r_{n-i+1} & \text{if } i \in \{2, 3, \dots, n-1\} \\ r_n & \text{if } i = n \end{cases} \text{ and } d'_i = \begin{cases} k & \text{if } i = 1 \\ k+2 & \text{if } i = 2 \\ 2 & \text{if } i = n-1 \\ d_n & \text{if } i = n \end{cases}$$

Lemma 4.4. In a $k, k+2$

Lemma 4.5. The flip operation will give you a valid arithmetical structure.

Proof. While the r values are changing position, d'_3 to d'_{n-2} stay the same because their neighbors are the same. However, we must show that d'_1, d'_2, d'_{n-1} , and d'_n are integers. Assuming $d_1 = k$ and $d_2 = k + 2$, we see that d'_1 is an integer since

$$d'_1 = \frac{kr'_2 + r'_n}{r'_1} = \frac{kr_{n-1} + r_n}{r_1 + r_{n-1} - r_2} = \frac{kr_{n-1} + kr_1 - kr_2}{r_{n-1} + r_1 - r_2} = k.$$

We know d'_2 is an integer because:

$$\begin{aligned} d'_2 &= \frac{kr'_1 + r'_3}{r'_2} = \frac{kr_1 + kr_{n-1} - kr_2 + r_{n-2}}{r_{n-1}} = \frac{kr_{n-1} + r_n + r_{n-2}}{r_{n-1}} \\ &= k + \frac{r_n + r_{n-2}}{r_{n-1}} = k + d_{n-1} = k + 2. \end{aligned}$$

And for d'_{n-1} , we see that

$$\begin{aligned} d'_{n-1} &= \frac{r'_{n-2} + r'_n}{r'_{n-1}} = \frac{r_3 + r_n}{r_2} = \frac{(k+2)r_2 - kr_1 + r_n}{r_2} = k+2 - \frac{kr_1 - r_n}{r_2} \\ &= k+2 - \frac{kr_1 - kr_1 + kr_2}{r_2} = k+2 - k = 2. \end{aligned}$$

And lastly, d'_n is

$$d'_n = \frac{r'_1 + r'_{n-1}}{r'_n} = \frac{r_1 + r_{n-1} - r_2 + r_2}{r_n} = \frac{r_1 + r_{n-1}}{r_n} = d_n.$$

□

5. ARITHMETICAL STRUCTURES ON CYCLES WITH A TRIPLE EDGE

From Proposition 4.2, we immediately get the following result for $\tilde{\mathcal{C}}_{n,3}$.

Proposition 5.1. *Let $(\mathbf{d}, \mathbf{r})$ be a non-Laplacian strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,3}$ with $r_1 > r_2$, then $d_1 = 2$ or $d_1 = 3$.*

Proposition 5.2. *Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,3}$ with $r_1 > r_2$ and $d_1 = 2$, then $5 \leq d_2 \leq 9$.*

Proof. Since $d_1 = 2$ then $r_1 = \frac{3r_2 + r_n}{2}$. It is known that $d_2 = \frac{3r_1 + r_3}{r_2}$. Substituting r_1 into this equation, we see that

$$d_2 = \frac{\frac{9r_2 + 3r_n}{2} + r_3}{r_2} = \frac{9r_2}{2r_2} + \frac{3r_n}{2r_2} + \frac{r_3}{r_2} = \frac{9}{2} + \frac{3r_n + 2r_3}{2r_2}.$$

Because d_2 must be an integer, the lowest value d_2 can be is 5. In order to evaluate the upper bound of d_2 , we split the known equation into four cases:

(1) Case 1: $r_3 < r_2, r_n < r_2$.

If $d_2 = 4.5 + \frac{3r_n}{2r_2} + \frac{r_3}{r_2}$ then $\frac{3r_n}{2r_2} = \frac{3}{2} * \frac{r_n}{r_2}$ where $\frac{r_n}{r_2} < 1$ and $\frac{r_3}{r_2} < 1$.

This means that $d_2 < 4.5 * \frac{3}{2}(1) + 1 = 4.5 + 1.5 + 1 = 7$.

So $d_2 < 7$ when $r_3 < r_2$ and $r_n < r_2$.

(2) Case 2: $r_3 < r_2, r_n \geq r_2$.

If $d_2 = 4.5 + \frac{3r_n}{2r_2} + \frac{r_3}{r_2}$ then $\frac{3r_n}{2r_2} = \frac{3}{2} * \frac{r_n}{r_2}$ where $\frac{r_n}{r_2} \geq 1$ and $\frac{r_3}{r_2} < 1$

but $r_n < r_1$ and $r_1 = \frac{3r_2 + r_n}{d_1}$ and $d_1 = 2$, so

$$r_n < \frac{3r_2 + r_n}{2}$$

$$2r_n < 3r_2 + r_n$$

$$r_n < 3r_2.$$

Then, when $r_n = 3r_2$,

$$d_2 < 4.5 + \frac{3}{2} * \frac{3r_2}{r_2} + \frac{r_3}{r_2}$$

$$d_2 < 4.5 + \frac{9}{2} + \frac{r_3}{r_2}$$

$$d_2 < 9 + \frac{r_3}{r_2}$$

where $\frac{r_3}{r_2} < 1$, so $d_2 < 10$ when $r_3 < r_2$ and $r_n \geq r_2$.

(3) Case 3: $r_3 > r_2, r_n \geq r_2$.

If $d_2 = 4.5 + \frac{3r_n}{2r_2} + \frac{r_3}{r_2}$ then $\frac{3r_n}{2r_2} = \frac{3}{2} * \frac{r_n}{r_2}$ where $\frac{r_n}{r_2} \geq 1$ and $\frac{r_3}{r_2} \geq 1$.

This means that $d_2 > 4.5 + \frac{3}{2} * (1) + 1 = 7$. This gives us a lower bound for d_2 when $r_3 > r_2$. In order to find the upper bound, we will be comparing the divisibility equations.

Let $d_1 = 2$. We will proceed using a proof by contradiction. Let $d_2 \geq 10$. Through divisibility equations, we know $d_n r_n = r_1 + r_{n-1}$. Since we are dealing with strongly smooth structures, we know that $d_1 \geq 2$, so we can use the following inequality: $2r_n \leq r_1 + r_{n-1}$. This can also be represented as $r_n - r_{n-1} \leq r_1 - r_n$. By using the rest of the divisibility equations, we can achieve the following inequality:

$$r_1 - r_n \geq r_n - r_{n-1} \geq r_{n-1} - r_{n-2} \geq \cdots \geq r_4 - r_3 \geq r_3 - r_2.$$

We can rewrite divisibility equation and plug in $d_2 = 10$ into $d_2 r_2 = 3r_1 + r_3$,

$$10r_2 - 3r_1 = r_3$$

$$9r_2 - 3r_1 = r_3 - r_2$$

$$3(3r_2 - r_1) = r_3 - r_2.$$

Now we rewrite another divisibility equation where we know $d_1 = 2$,

$$d_1 r_1 = 3r_2 + r_n$$

$$2r_1 = 3r_2 + r_n$$

$$r_1 - r_n = 3r_2 - r_1.$$

Because of this equality above, we know that $3(3r_2 - r_1) = 3(r_1 - r_n)$. According to our inequality above, $r_1 - r_n \geq r_3 - r_2 = 3(r_1 - r_n)$. But $r_1 - r_n \not\geq 3(r_1 - r_n)$ since r_1 and r_n are positive integers, which is a contradiction. So, $d_2 < 10$ when $r_3 > r_2$.

(4) Case 4: $r_3 > r_2, r_n < r_2$.

We know that in most cases $r_2 > r_3$, but when $r_3 > r_2$, then [more to say here but I can add it](#) $r_{n+1} > r_n$ in order to keep the structure smooth. With this rule, $r_n \not\leq r_2$. So this case does not exist. $\square$

Proposition 5.3. *Let $(\mathbf{d}, \mathbf{r})$ be a strongly smooth arithmetical structure on $\tilde{\mathcal{C}}_{n,3}$ with $r_1 > r_2$. When $d_1 = 3$, we have that $4 \leq d_2 \leq 5$.*

Proof. Let $d_1 = 3$. Then, $r_1 = \frac{3r_2+r_n}{3}$. It is known that $d_2 = \frac{3r_1+r_3}{r_2}$. Substituting r_1 , we see that

$$d_2 = \frac{3(\frac{3r_2+r_n}{3}) + r_3}{r_2} = \frac{3r_2 + r_n + r_3}{r_2} = 3 + \frac{r_n + r_3}{r_2}.$$

Since $\frac{r_n+r_3}{r_2}$ is positive and $d_2 \in \mathbb{N}$, then $d_2 \geq 4$.

For the upper bound, we will proceed using a proof by contradiction. Thus, $d_2 \geq 6$. Using divisibility conditions, we know that $d_n = \frac{r_1+r_{n-1}}{r_n}$. Therefore, $d_n r_n = r_1 + r_{n-1}$. Because we are assuming strong smoothness, we can rewrite this equation as an inequality, where $2r_n \leq r_1 + r_{n-1}$. This can also be represented as $r_n - r_{n-1} \leq r_1 - r_n$. If you keep using these divisibility equations, you get an inequality that looks like

$$r_1 - r_n \geq r_n - r_{n-1} \geq r_{n-1} - r_{n-2} \geq \cdots \geq r_4 - r_3 \geq r_3 - r_2.$$

Using divisibility conditions we know that, $d_2 r_2 = 3r_1 + r_3$. Solving for $r_3 - r_2$, we get

$$r_3 - r_2 = (d_2 - 1)r_2 - 3r_1 \geq 5r_2 - 3r_1.$$

Using the divisibility condition $d_1 = \frac{3r_2+r_n}{r_1}$, we get $3r_1 = 3r_2 + r_n$, we know that $r_2 = \frac{3r_1-r_n}{3}$. Plugging this into $5r_2 - 3r_1$, we see that

$$5r_2 - 3r_1 = 2r_1 - \frac{5}{3}r_n \leq r_3 - r_2$$

. Therefore, $r_1 - r_n \geq 2r_1 - \frac{5}{3}r_n$, which simplifies to $-r_1 - \frac{8}{3}r_n \geq 0$, a contradiction. Thus, when $d_1 = 3$, $d_2 \leq 5$. $\square$

[There is more to explain in the above result...](#)

Theorem 5.1. *All strongly smooth graphs $\tilde{\mathcal{C}}_{n,k}$ where $d_1 = 3$ and $d_2 = 4$ originate from a smaller weakly or strongly smooth graph where $d_1 = 1$ or 2 .*

Is this true? Do they all originate from smaller strongly smooth where $d_1 = 2$? Some originate from Laplacians, right? Do some of them originate from something else? Do we have data about this? Can we get data?